

# Mobile Communications

Slide set 1: Wave propagation and description of mobile radio channels

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# About this course

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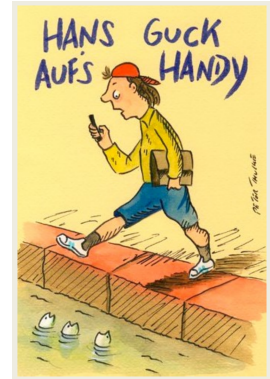
- **Instructors**

- ▶ Dr.-Ing. Omar Alnaseri

- **Scope:** approx. 33 h of lectures and exercises

- **Contact**

- ▶ `alnaseri.omar@dozent.dhbw-ravensburg.de`



# Overview of the contents (1) — Fundamentals

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## Slide set 1 — Wave propagation and description of mobile radio channels

- Free space propagation
- Mobility
- Multipath propagation and propagation effects (reflection, scattering, diffraction)
- Fading
- Calculating with decibels
- Link budgets

## Slide set 2 — Channel models and radio network planning

- Radio channel: frequency selectivity, time variance
- Basic principles of MIMO
- Wave propagation along railway lines / Doppler effect
- Modelling of mobile radio channels
- Radio network planning: reducing interference, increasing capacity, increasing received power

# Overview of the contents (2) — State of the art

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## Slide set 3 — Physical layer, example GSM

- Overview of mobile communication standards
- Parameters of the physical layer
- Frame structure, bursts, physical channels
- Signal processing: source coding, channel coding (block and convolutional codes), interleaving
- Power control, channel estimation and equalisation
- Logical channels; GSM network architecture

## Slide set 4 — Digital modulation

- Advantages of digital modulation, applications
- Modulation with a sinusoidal carrier
- I-Q modulator and demodulator; meaning of I and Q
- ASK, PSK and variants (QPSK, OQPSK,  $\pi/4$ -DQPSK, MSK, GMSK, 8-PSK)
- QAM, FSK

**Not everything here is examinable — we will say clearly what is.**



# Overview of the contents (3) — Current & future

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## Slide set 5 — LTE, 5G and beyond

### ■ LTE

- ▶ Overview of the LTE network architecture
- ▶ Advantages
- ▶ Coverage in Germany

### ■ 5G and new standards

- ▶ Challenges with new standards
- ▶ Research on future cellular technologies

### ■ 5G in detail

- ▶ Requirements and applications
- ▶ Network architecture and special features
- ▶ 5G channel model
- ▶ Network expansion (Netzausbau) and coverage in Germany

### ■ The way to 6G

# Overview of the contents (4) — Satellites and AI

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## Slide set 6 — Radio communication with satellites

- Special features of satellite communication
- Structure of a satellite: TT&C subsystem, communication subsystem
- Ground station; design of a link budget
- Trends in satellite-based mobile communication
  - ▶ OneWeb constellation and similar projects
  - ▶ E-band radio communication
  - ▶ Laser Communication Terminals (LCT)

## Slide set 7 — Artificial Intelligence

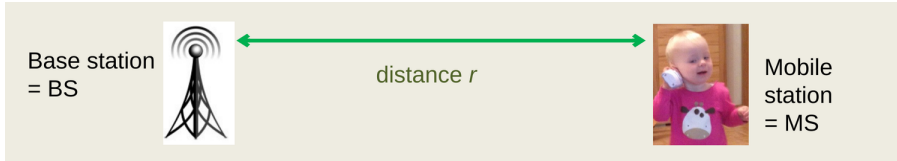
- Overview of AI and its applications
- Focus on AI in communication systems

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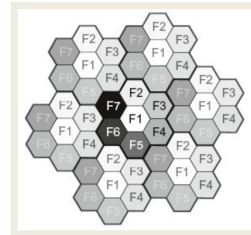
# Wave propagation and description of mobile radio channels

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# Basic idea of a mobile radio network



- Both BS and MS can **transmit and receive** ( $T_x$  = transmitter,  $R_x$  = receiver)
- Received power falls as the distance  $r$  grows  
 $\Rightarrow$  signal quality decreases  $\Rightarrow$  **finite range**
- Nationwide coverage  $\Rightarrow$  **cellular structure**
  - ▶ a new BS after a certain distance
  - ▶ each BS serves one *radio cell*
  - ▶ neighbouring cells use different frequencies to avoid interference



# What does mobility mean?

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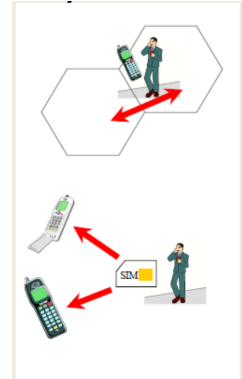
## ■ Terminal mobility

- ▶ the terminal device moves within the service area
- ▶ accessibility at any location

## ■ Personal mobility (mobility of the subscriber)

- ▶ independent of the terminal device (e.g. by changing the SIM card)
- ▶ use of the subscribed telecommunication services
- ▶ reachable under the *same* number

Source: uni.paniladen.de — Grundlagen des Mobilfunks



# Mobility in various radio networks

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- **In the home**: e.g. cordless telephone; range typically 50 – 200 m
- **Regional / local**: e.g. company networks, private radio systems with a few cells
- **National**: radio network limited to one country (e.g. the former analogue C-network)
- **International**: e.g. the GSM network
- **Global** (anywhere on earth, also at sea): satellite-based mobile radio communication

# Prerequisites for mobility

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- **Radio coverage** in the desired area
  - ▶ achieved by radio network planning, using suitable wave propagation models — e.g. taking the terrain into account
- **Adaptive assignment** of the mobile station to a radio cell
  - ▶ a cell change procedure is necessary  $\Rightarrow$  **handover**
- **Mobility management**, which provides
  - ▶ automatic paging of the subscriber for an incoming call or data
  - ▶ roaming

# Mobility: why a handover is needed

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- A handover becomes necessary, for example, when
  - ▶ the **reception level  $C$  is too low** — the MS has moved too far from its current base station
  - ▶ there is **co-channel interference**: frequencies are reused after a certain distance, so even with a high level  $C$  the ratio  $C/I$  (carrier-to-interference) can be low  $\Rightarrow$  **high bit error rate**
  - ▶ there is **too much traffic** in the cell
- Elementary requirement for every handover:
  - ▶ the user must **not notice it** — no interruption of the call or of the data session



# Mobility: hard handover and soft handover

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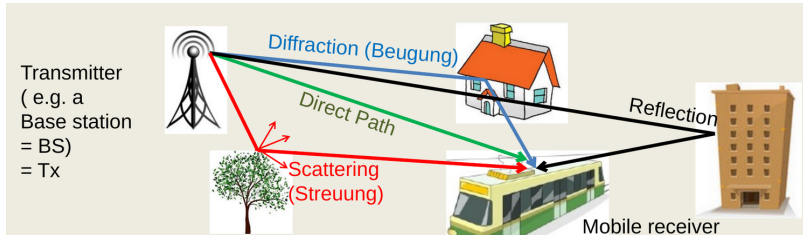
- **Hard handover** — “break before make”

- ▶ the existing connection to the current cell is disconnected *before* the connection to the new cell is established
- ▶ GSM neighbouring cells use different radio frequencies, therefore **GSM handovers are always hard handovers**

- **Soft handover** — “make before break”

- ▶ the connection to the new cell is established *before* the old one is released
  - ▶ for a short time there are two simultaneous connections to different cells
  - ▶ used in UMTS, no longer used in LTE
- further reading: [3glteinfo.com/soft-handover-lte](http://3glteinfo.com/soft-handover-lte)

# Multipath propagation for radio signals



- **Direct path** — only exists under **Line of Sight** (LOS) conditions
- With polarisation matching and no obstacles the received power of the direct path follows the **Friis transfer equation** (*Friis'sche Übertragungsgleichung*, free-space propagation)

$$P_{Rx} = P_{Tx} \cdot G_{Tx,LOS} \cdot \left( \frac{c_0}{4\pi \cdot r \cdot f} \right)^2 \cdot G_{Rx,LOS}$$

decrease  $\sim 1/r^2$ ;  $P_{Tx}$ : emitted Tx power

With multipath propagation: **superposition** of many dispersion paths, arriving **time-delayed** from different directions.

# Interactive lab 1 — Multipath propagation & the Friis equation

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## What you can explore

The notebook redraws the scene from the previous slide — direct path, diffraction over a house, scattering at a tree, reflection from a building — and computes the received power live.

**Sliders:**  $P_{Tx}$ , frequency, distance  $r$ , antenna gains, and the loss of each individual multipath component.

### Try this:

- Double the distance  $r$  — by how many dB does  $P_{Rx}$  drop? Does it match the  $1/r^2$  law?
- Move the frequency from 700 MHz to 6 GHz at constant  $r$ . Compare with slides 17–19.
- Switch the multipath components on and off: when does the *total* received power exceed the LOS power alone?



[Colab notebook 01](#)

[Multipath & Friis](#)

Runs in Google Colab  
— no installation needed.

# Where does the Friis transfer equation come from?

**A. Modeling of Interference Power**

The approach presented below assumes free space propagation between a Tx and Rx where the Tx radiates a power  $P_{Tx}$  at a frequency  $f$  and the victim Rx receives the signal in-band as an interference signal. Figure 3 shows a general constellation involving Tx, Rx, antenna patterns and the definition of elevation angles towards the LoS path.

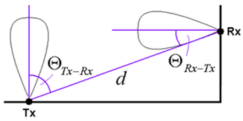


Figure 3. General definition of angles between transmitter and receiver

If the antenna patterns are also dependent on the azimuth angle, azimuth has to be considered as well.

In general, the power density  $S$  (in  $W/m^2$ ) at a victim Rx which has been generated by a Tx antenna can be determined as

$$S = \frac{P_{Tx} \cdot G_{Tx}(\Theta_{Tx-Rx}, \varphi_{Tx-Rx})}{4\pi \cdot d^2} \quad (1)$$

where  $P_{Tx}$  is the total transmitted Power,  $G_{Tx}(\Theta_{Tx-Rx}, \varphi_{Tx-Rx})$  is the Tx antenna gain at the considered frequency  $f$  in the direction of the Rx (direction described by the elevation

angle  $\Theta_{Tx-Rx}$  and the azimuth angle  $\varphi_{Tx-Rx}$ ), and  $d$  is the distance between Tx and Rx.

The Rx antenna will suffer interference from the incident signal. The received interference power  $P_{Rx}$  at a distance of  $d$  is

$$P_{Rx} = S \cdot A_{eff} = \frac{P_{Tx} \cdot G_{Tx}(\Theta_{Tx-Rx}, \varphi_{Tx-Rx})}{4\pi \cdot d^2} \cdot \frac{G_{Rx}(\Theta_{Rx-Tx}, \varphi_{Rx-Tx}) \cdot c_0^2}{4\pi \cdot f^2} \quad (2)$$

where  $G_{Rx}(\Theta_{Rx-Tx}, \varphi_{Rx-Tx})$  is the antenna gain of the victim Rx towards the Tx,  $f$  is the Rx frequency and  $c_0$  the speed of light in vacuum. This equation describes free space propagation and is known as Friis equation. It assumes that the Rx is positioned in the far field of the transmitter: For antennas physically larger than  $\lambda/2$  (where  $\lambda$  is radiated wavelength), the Rx is in the far field of the Tx for  $d > d_f = 2D^2/\lambda$  (far field condition). The parameter  $D$  corresponds to the physical length of an antenna, or the diameter of a 'dish' antenna. In addition, the following conditions have to be fulfilled:  $d_f \gg D$  and  $d_f \gg \lambda$ .

The received power may be attenuated due to harness losses between the Rx antenna and the receiver input (e.g. 2 dB losses). This can be respected by a factor  $L_{RM}$  ( $\leq 1$ ) in the equation. The interference power at the receiver input is then

$$P_{Rx} = L_{RM} \cdot \frac{P_{Tx} \cdot G_{Tx}(\Theta_{Tx-Rx}, \varphi_{Tx-Rx})}{4\pi \cdot d^2} \cdot \frac{G_{Rx}(\Theta_{Rx-Tx}, \varphi_{Rx-Tx}) \cdot c_0^2}{4\pi \cdot f^2} \quad (3)$$

Extract from: J. Timmermann, C. Imhof, D. Leberherz, J. Lange, "Ensuring Radio Frequency Compatibility (RFC) on-Board a Satellite by Early Analysis and Efficient Methods for Field Prediction", *International Journal on Advances in Telecommunications*, vol. 8, no. 3 & 4, 2015.

# Free space attenuation — attenuation in decibel (dB)

## Friis transfer equation

$$P_{Rx} = P_{Tx} \cdot G_{Tx,LOS} \cdot \underbrace{\left( \frac{c_0}{4\pi \cdot r \cdot f} \right)^2}_{\text{free space attenuation}} \cdot G_{Rx,LOS}$$

a damping factor  $> 0$  and  $< 1$

- Attenuation in decibel (dB): apply  $10 \log_{10}$  and take the absolute value (*Betrag*)

$$\left| 10 \cdot \log_{10} \left( \frac{c_0}{4\pi \cdot r \cdot f} \right)^2 \right| \quad \Rightarrow \quad \text{e.g. a value of } \mathbf{90}, \text{ i.e. 90 dB of free space attenuation}$$

## Exercise 8 — Free space propagation (*Freiraumausbreitung*)

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### Task

A transmitter transmits at a frequency of **1800 MHz**. Free space propagation is considered.

- a) Calculate the free space attenuation in dB at a distance of **100 m** and of **10 km**.
- b) How do the values from a) change when the frequency is **halved**?

$$a_{\text{FS}}[\text{dB}] = \left| 10 \log_{10} \left( \frac{c_0}{4\pi \cdot r \cdot f} \right)^2 \right| = 20 \log_{10} \left( \frac{4\pi \cdot r \cdot f}{c_0} \right)$$

*Übungsblatt Wellenausbreitung, Aufgabe 8*

# Relevance of the Friis equation in mobile communications (1)

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According to [antenna-theory.com/basics/friis.php](http://antenna-theory.com/basics/friis.php)

- “The Friis equation shows that **more power is lost at higher frequencies**. The difference between the power received and the power transmitted is known as **path loss**. Said differently: the path loss is higher for higher frequencies.”
- “This is why mobile phones generally operate **below 2 GHz**. There may be more spectrum available at higher frequencies, but the associated path loss will not enable quality reception.”
- “At 60 GHz — the millimetre-wave region — the path loss is very high, so only point-to-point communication is possible: receiver and transmitter in the same room, facing each other.”

# Relevance of the Friis equation in mobile communications (2)

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According to [antenna-theory.com/basics/friis.php](http://antenna-theory.com/basics/friis.php)

- “Do you think the mobile phone operators are happy about the LTE (4G) band that operates at **700 MHz**? The answer is yes: this is a lower frequency than antennas traditionally operate at, so from the Friis equation the **path loss will be lower** as well. Hence they can *cover more ground* — which is why this is called *high quality spectrum*.”
- “*Side note: on the other hand, the phone makers now have to fit an antenna for a larger wavelength into a compact device, so the antenna designer’s job got a little more complicated!*”

$$\lambda = \frac{c}{f}$$

smaller frequency  $\Rightarrow$  larger wavelength  $\Rightarrow$  **larger antenna required**



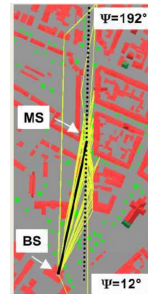
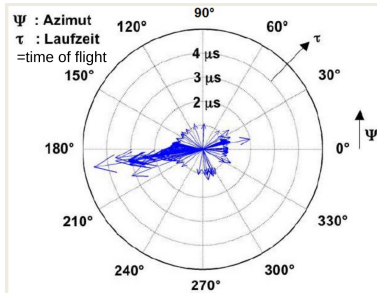
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... now some details  
on multipath propagation

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# Multipath propagation — practical example: paths arriving in Karlsruhe

Simulated transmission paths: angular distribution of the paths arriving at the mobile station (DoA = **D**irection **o**f **A**rrival). Transmitter (BS) on top of a high building, receiver (MS) in a street canyon (*Straßenschlucht*).



**Waveguide effect  $\Rightarrow$  two main dominant directions**

# Details on reflection

- For an **ideally smooth** surface the law of reflection applies: angle of incidence = angle of reflection (coherent component in the direction of reflection)
- Real surfaces are **rough** to some extent
- One speaks of *reflection* if the **Fraunhofer criterion** is fulfilled, i.e. the surface roughness is weak:

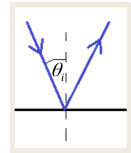
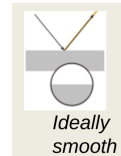
$$\sigma < \frac{\lambda_0}{32 \cdot \cos \theta_i}$$

$\sigma$ : standard deviation of the surface height variation

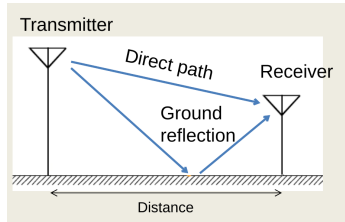
$\lambda_0$ : wavelength of the incident wave

$\theta_i$ : angle of incidence

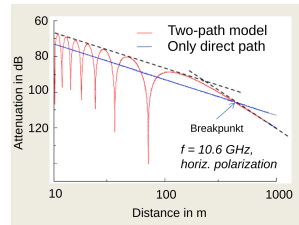
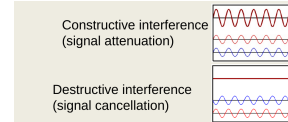
- The equation shows:
  - ▶ for  $\theta_i \rightarrow 90^\circ$  we get  $\sigma \rightarrow \infty$ : the **flatter** (*streifender*) the incidence, the larger the roughness that still gives reflection
  - ▶ the **larger the wavelength** (the lower the frequency), the greater the  $\sigma$  that can be accepted



# Reflection: modelling with the two-path model



- Constructive and destructive interference are both possible, depending on the Rx position
- $\Rightarrow$  **spatial variation** of the received signal
- $\Rightarrow$  **fading** (*Schwund*) as a consequence of the superposition of paths



# Interactive lab 2 — Reflection and the two-path model

## What you can explore

Direct wave and ground-reflected wave are added *as fields*, with phase. Three views: the geometry with the phasors, the received power versus distance, and an animation of the two waves superposing.

$$\Delta\phi = \frac{2\pi}{\lambda}(d_2 - d_1) + \pi, \quad \text{breakpoint } d_{BP} = \frac{4 h_{Tx} h_{Rx}}{\lambda}$$

Try this:

- Find a distance where the two waves cancel. How far do you have to move to get the signal back?
- Read the slope before and after  $d_{BP}$ : 20 dB/decade → **40 dB/decade**.
- Raise the Tx antenna: what happens to  $d_{BP}$  — and why does that matter for the cell radius?



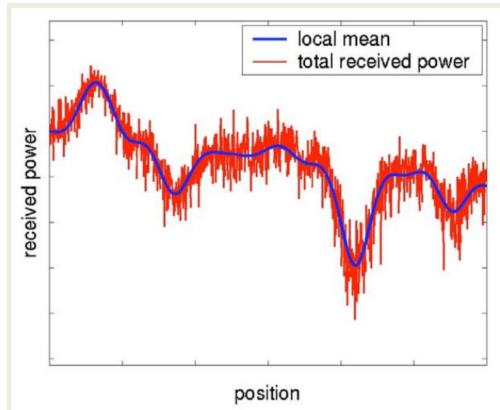
[Colab notebook 03](#)

[Reflection, two-path](#)

Runs in Google Colab  
— no installation needed.

# Visualisation of fading

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Reference: T. Zwick et al., “Radio Wave Propagation Fundamentals”, Institut für Hochfrequenztechnik und Elektronik, Universität Karlsruhe.

# Details on fading: fast fading

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- The multipath scenario is **frequency-dependent**, because reflection, scattering and diffraction are themselves frequency-dependent  
⇒ the radio channel is **frequency-selective**
- The radio channel is also **time-variant**. Causes:
  - ▶ propagation times and phases of the multipath signals change with time, e.g. through moving scatterers (passing birds, moving cars)  
⇒ constructive or destructive interference ⇒ rapid variations of the received level *even if  $T_x$  and  $R_x$  do not move*
  - ▶ a moving receiver travels through a spatial interference pattern  
⇒ also rapid temporal fluctuations of the received power

Rapid temporal fluctuations of the received signal = **fast fading**  
⇒ the *number* of paths arriving at the receiver stays constant over time

# Details on fading: slow fading

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- If the **number** of paths changes slowly over time (new paths appear, old paths disappear): the **mean value** of the received signal changes  $\Rightarrow$  **slow fading**
- Causes, for example:
  - ▶ the receiver moves into a shadowed area and the LOS path disappears
  - ▶ slowly changing weather conditions
  - ▶ driving into a different propagation situation
- In reality: **superposition of fast fading and slow fading**

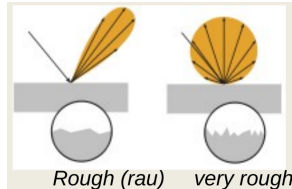
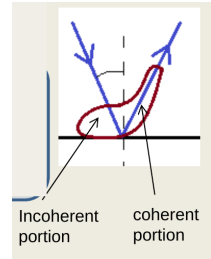
The distances over which slow fading becomes noticeable are much larger than the period of the superimposed interference pattern. Therefore slow fading is also called **large scale fading**, and fast fading **small scale fading**  $\Rightarrow$  the fluctuations in slow fading are much smaller than in fast fading.



# Details on scattering (1)

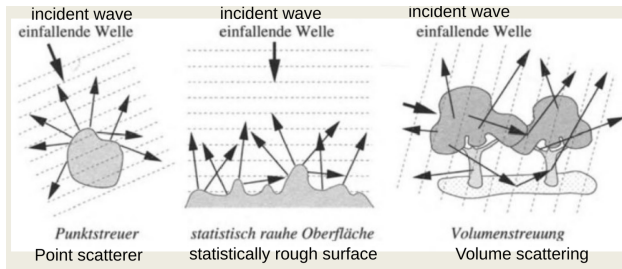
Scattering means that the power is distributed into **different directions** after the interaction with a surface:

- in the direction of reflection  $\Rightarrow$  **coherent portion**
- in all other directions  $\Rightarrow$  **incoherent portion** (= diffuse portion)
- The interaction with a surface is called *scattering* when the **Fraunhofer criterion is no longer fulfilled**, i.e. the surface roughness is higher than the criterion permits.



## Details on scattering (2)

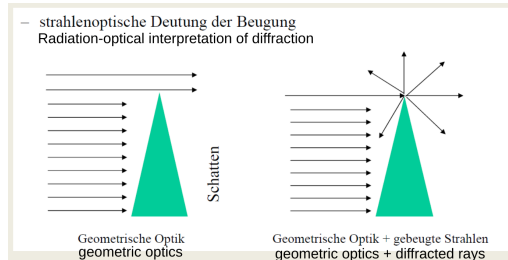
- There is not only scattering from statistically rough *surfaces*, but also scattering from **objects**.
- The smaller the ratio of wavelength to object dimension, the larger the **incoherent portion**, i.e. the more scattering.



- Calculation of the scattering (polarimetric scattering matrix) is possible with numerical or analytical methods; numerical: Kirchhoff theory for the coherent part, Kirchhoff theory or the Small-Perturbation Method for the incoherent part (if the surface has jumps).

# Details on diffraction (*Beugung*) (1)

- In **geometric optics** a source emits “rays”. If they meet an obstacle, a *geometric shadow* is created — an Rx in the shadow receives nothing.
- For infinitely high frequencies this is correct: the field strength in the shadow really does become zero.
- Reality at finite frequencies (e.g. the GHz range of mobile radio): **diffraction** — there *is* a signal in the geometric shadow.
  - ▶ diffraction is also observable for light (THz range)  $\Rightarrow$  the double-slit experiment (*Doppelspalt-Versuch*)



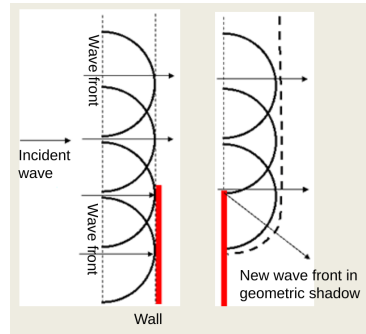
Source: T. Kürner, *Planung terrestrischer Netze*, TU Braunschweig, 2009.

# Details on diffraction (2): Huygens' principle

Cause of diffraction — **Huygens' principle**:

“Every point of a wave front is the starting point of a new spherical wave.”

- This creates, for example, a path into the geometric shadow (reaching around obstacles) — but the signal there is **strongly attenuated**
- $t = 0$ : the wave front just reaches the wall; the region behind is still shadowed
- $t > 0$ : new spherical waves form at the edge and propagate into the shadow region



# Interactive lab 3 — Huygens' principle and diffraction

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## What you can explore

Three visualisations: the *wavelet construction* (how the envelope of the secondary waves becomes the new wave front), *diffraction at a wall* for  $t = 0$  and  $t > 0$ , and an interactive explorer for wavelength, aperture and time.

### Try this:

- Increase the wavelength: how much further does the wave reach into the geometric shadow?
- Compare a 700 MHz and a 3.5 GHz setting — which one gets around the building, and which 5G band does that explain?
- Reduce the aperture until it is smaller than  $\lambda$ : what is left of the “straight ray” picture?



[Colab notebook 02](#)

[Huygens, diffraction](#)

Runs in Google Colab  
— no installation needed.

# Diffraction: some calculation methods I

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- **Knife-edge diffraction** (details on the following slides)
- **High-frequency approximations**
  - ▶ **GTD** — Geometrical Theory of Diffraction:  
an extension of geometrical optics — besides direct and reflected rays there are also *diffracted rays* (edge and corner diffracted rays, and surface creeping rays); prerequisite: all objects in the far field
  - ▶ **UTD** — Uniform Theory of Diffraction:  
an improved version of GTD with correction terms; gives more realistic results at the shadow edge. The UTD solution is a high-frequency approximation for *canonical* objects:
    - infinitely long metallic wedge (*unendlich langer metallischer Keil*)
    - infinitely long metallic double wedge (*Doppelkeil*)
    - infinitely long metallic convex surface (curved outwards)
    - infinitely long metallic polygonal cylinder
- General objects are composed of canonical objects: compute a solution for each and assemble the solutions.
- **Field-theoretical methods** (Parabolic Equation Method)
- **Empirical methods**

Translated from: `tuprints.ulb.tu-darmstadt.de/946/1/Dissertation_Andreas_Paech`

# Diffraction: some calculation methods II

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## The difference between GTD and UTD

Although GTD can compute the diffracted fields in the shadow area, there is still a **jump in the field strength at the shadow edge**. The reason: the direct rays that just pass the edge are, for example, plane waves with constant amplitude, whereas the diffracted cylindrical waves decay as  $1/\sqrt{R}$ . In this transition region GTD does not give correct fields.

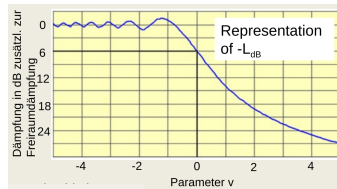
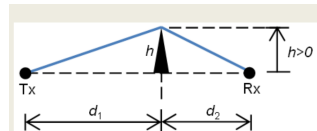
This is remedied by extending GTD: the diffracted cylindrical waves are adjusted near the shadow boundaries by **correction terms**, so that the resulting total field of direct, reflected and diffracted contributions is **continuous**. This extension is called the **Uniform Theory of Diffraction (UTD)**.

# Diffraction: modelling with the knife-edge approach

- Replace the obstacle by an **absorbing half-plane** (“knife-edge”) of height  $h$  between Tx and Rx
- Introduce the **Fresnel–Kirchhoff parameter**  $\nu$  (Greek letter “nü”):

$$\nu = h \cdot \sqrt{\frac{2}{\lambda} \cdot \left( \frac{1}{d_1} + \frac{1}{d_2} \right)}$$

⇒ **the attenuation increases with greater height  $h$  or with higher frequency**



Approximation of the additional attenuation (on top of the free space attenuation):

$$L_{dB} = \begin{cases} -(6 + 9 \cdot \nu - 1.27 \nu^2) & 0 \leq \nu \leq 2.4 \\ -(13 + 20 \cdot \log_{10}(\nu)) & \nu > 2.4 \end{cases}$$

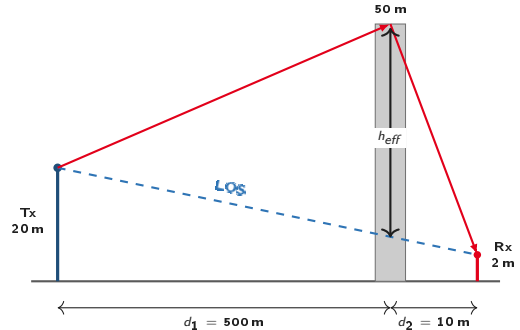


## Exercise 9 — Knife-edge diffraction (*Knife edge Beugung*)

### Task

A transmitter site of height **20 m** is **500 m** away from a high-rise building of height **50 m**. A vehicle drives behind the building at a distance of **10 m**, antenna height **2 m**. The driver wants a **GSM 1800 MHz** connection.

- Sketch the scenario and calculate the attenuation *in addition* to the free space attenuation caused by diffraction.
- Without the building a transmitter with **1 mW** would be sufficient. Which transmit power is needed because of the building?



$$\nu = h \sqrt{\frac{2}{\lambda} \left( \frac{1}{d_1} + \frac{1}{d_2} \right)}, \quad L_{dB} = -(13 + 20 \log_{10} \nu) \text{ for } \nu > 2.4$$

Übungsblatt Wellenausbreitung, Aufgabe 9

# Practical example: the knife-edge approach on a satellite

- A transmitter transmits towards earth; an instrument on the same satellite sees the **interfering signal**
- Ensuring **Radio Frequency Compatibility (RFC)**: minimise the interfering signal by inserting a **metal wall**
- Calculate the delta attenuation by two methods:

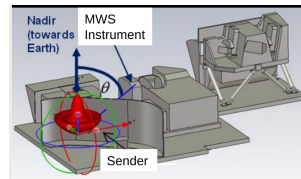


## Method 1: field simulation

- simulate the field strength with and without the wall
- attenuation = the difference; disadvantage: high computing effort

## Method 2: knife-edge approach

- additionally consider the antenna gain by a correction factor



J. Timmermann, C. Imhof, D. Leberer, J. Lange, "Ensuring Radio Frequency Compatibility (RFC) on-Board a Satellite...", *Int. Journal on Advances in Telecommunications*, vol. 8, no. 3 & 4, pp. 152–161, 2015. — Same authors, "Application of Knife-Edge Diffraction Theory to Optimize Radio Frequency Compatibility On-Board a Satellite", SPACOMM, Barcelona, April 2015.

[http://personales.upv.es/thinkmind/dl/journals/tele/tele\\_v8\\_n34\\_2015/tele\\_v8\\_n34\\_2015\\_4.pdf](http://personales.upv.es/thinkmind/dl/journals/tele/tele_v8_n34_2015/tele_v8_n34_2015_4.pdf)

*International Journal on Advances in Telecommunications, vol 8 no 3 & 4, year 2015, <http://www.iariajournals.org/telecommunications/>*

## **Ensuring Radio Frequency Compatibility (RFC) on-Board a Satellite by Early Analysis and Efficient Methods for Field Prediction**

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**Abstract**—Transmitters (=Tx) on-board a satellite generate an electromagnetic environment with potential impact on victim receivers (=Rx, e.g., instruments) placed nearby. Ensuring Radio Frequency Compatibility (RFC) on-board a satellite is hence an important point to be considered during satellite design and requires an optimized satellite configuration. This contribution concentrates on RFC issues in practical satellite design by considering the future MetOp-SG meteorological satellites: First, an overview is given summarizing the various transmitters and instrument receivers on-board the satellites. Then, the fundamentals of RFC analysis are presented showing the method how to compute the coupling factor between a Tx and a victim Rx. To improve the decoupling, MetOp-SG satellites are housing dedicated baffles between Tx and Rx antennas. Therefore, the contribution finally studies in detail

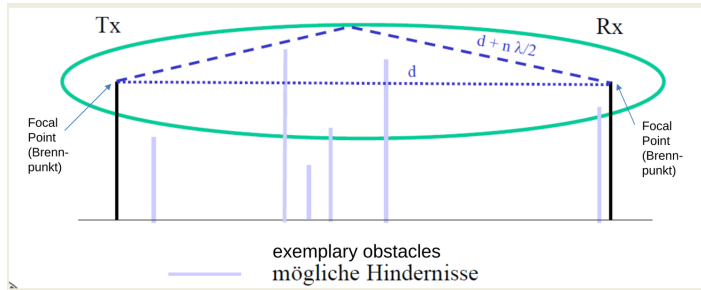
a later project phase. The approach is hence to define a preliminary configuration and to run an RFC analysis which investigates the coupling between critical Tx and Rx combinations. In an early project phase, the unintended signal at a victim receiver shall be well below (typically 20 dB) the maximum acceptable value, whereas the difference is called RFC margin. On the other side, a satellite configuration will not only be optimized w.r.t. RFC. Other aspects (such as center of mass, minimization of harness length etc.) have to be taken into account and will lead to some configuration changes. In the end, a compromise will be required ensuring positive margins in all considered disciplines.

As the optimization exercise is typically not finished in

# Diffraction: Fresnel zones

## Definition

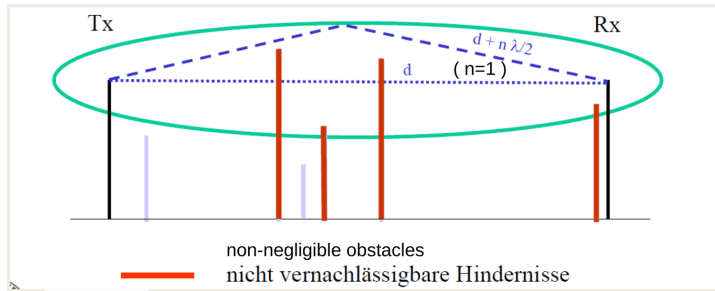
The  $n$ -th Fresnel zone is a 3D ellipsoid in space in which the sum of the distances to the focal points Tx and Rx is greater by  $n \cdot \frac{\lambda}{2}$  than the distance  $d$  between the focal points.



Source: T. Kürner, *Planung terrestrischer Netze*, TU Braunschweig, 2009.

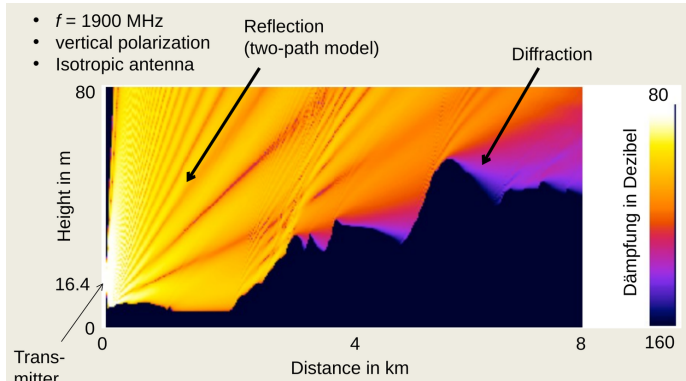
# Diffraction: relevant diffractive obstacles in radio transmission

- If an obstacle **does not reach into the first Fresnel zone**, its influence *with regard to diffraction* can be neglected to a good approximation.
- **This does not generally apply** to the influence of scattering and reflection.



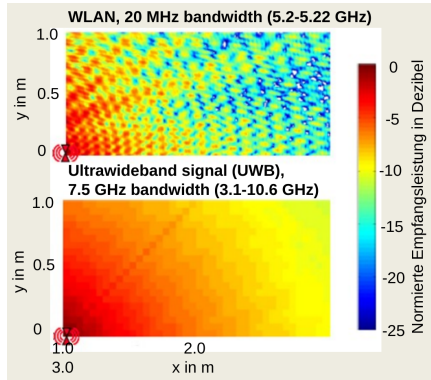
Source: T. Kürner, *Planung terrestrischer Netze*, TU Braunschweig, 2009.

# Practical example: signal attenuation in hilly terrain



The greater the diffraction angle (from the LOS towards the Rx in the shadow region), the greater the attenuation.

# Practical example: signal attenuation for WLAN and UWB (indoor)



## Influence of bandwidth:

The broader the bandwidth of the signal, the less likely the *simultaneous* destructive interference on all signal frequencies — i.e. the **smaller the spatial signal fluctuations**.

Reference: course material of the European School of Antennas / Course on Ultra Wideband Antennas, 3–7 April 2006, Universität Karlsruhe.  
Assumption: indoor environment, isotropic Tx and Rx antenna, simulated by ray tracing.

---

## Calculating with decibels

---



# Friis equation in logarithmic representation (1)

---

$$P_{Rx} = P_{Tx} \cdot G_{Tx,LOS} \cdot \left( \frac{c_0}{4\pi \cdot r \cdot f} \right)^2 \cdot G_{Rx,LOS}$$

$$P_{Rx,dB} = 10 \log_{10} P_{Rx} = 10 \log_{10} \left\{ P_{Tx} \cdot G_{Tx,LOS} \cdot \left( \frac{c_0}{4\pi \cdot r \cdot f} \right)^2 \cdot G_{Rx,LOS} \right\}$$

Exploiting  $\log(a \cdot b) = \log a + \log b$  yields:

$$\begin{aligned} P_{Rx,dBW} &= 10 \log_{10} P_{Rx}[\text{W}] \\ &= 10 \log_{10} P_{Tx}[\text{W}] + 10 \log_{10} G_{Tx,LOS} + 10 \log_{10} \left( \frac{c_0}{4\pi \cdot r \cdot f} \right)^2 + 10 \log_{10} G_{Rx,LOS} \\ &= P_{Tx,dBW} + G_{Tx,LoS,dBi} + 10 \log_{10} \left( \frac{c_0}{4\pi \cdot r \cdot f} \right)^2 + G_{Rx,LoS,dBi} \end{aligned}$$

“dB” indicates a decibel quantity — details on the next slide

# Friis equation in logarithmic representation (2) — power and “dB”

## Power and “dB”

- $P$  in **Watt**  $\Rightarrow$  applying  $10 \log_{10}$  gives a value in **dBW**
- $P$  in **mW**  $\Rightarrow$  applying  $10 \log_{10}$  gives a value in dBmW, in short notation **dBm**

$$\begin{aligned} P_{dBW} = 10 \log_{10}(P[\text{W}]) &\iff P[\text{W}] = 10^{\frac{P_{dBW}}{10}} \\ P_{dBm} = 10 \log_{10}(P[\text{mW}]) &\iff P[\text{mW}] = 10^{\frac{P_{dBm}}{10}} \end{aligned}$$

## Antenna gain and “dB”

- The gain  $G$  is angle-dependent; applying  $10 \log_{10}$  to  $G$  gives the value in **dB**i — the “i” says that the reference is an *isotropic* (spherical) radiator

$$G_{dB_i} = 10 \log_{10}(G) \iff G = 10^{\frac{G_{dB_i}}{10}}$$

$G > 1$  (i.e.  $> 0$  dB*i*): the radiation is **focussed** at the considered angle, compared with the isotropic antenna.

$G < 1$  (i.e.  $< 0$  dB*i*): **less** radiation than with the isotropic antenna  $\Rightarrow$  a negative value in dB*i*.

# Exercises 1, 2 and 5 — dBm, mW and the 30 dB rule

---

## Exercise 1 — into dBm

Convert a transmitting power of **13 W** into dBm.

## Exercise 2 — into mW

Convert **0 dBm** into mW.

## Exercise 5 — dBW vs. dBm

Prove the relationship

$$P[\text{dBW}] = P[\text{dBm}] - 30.$$

*Übungsblatt Wellenausbreitung, Aufgaben 1, 2 und 5*

# Friis equation and dB

---

- If values are given in dB and the Friis equation is to be used, one possible strategy is:
  - ▶ convert the dB values into **linear** values and insert the linear values into the Friis equation

$$P_{Rx} = P_{Tx} \cdot G_{Tx,LOS} \cdot \left( \frac{c_0}{4\pi \cdot r \cdot f} \right)^2 \cdot G_{Rx,LOS}$$

- Result required in **dBW**: take the received power in *Watt* and apply  $10 \log_{10}$
- Result required in **dBm**: take the received power in *mW* and apply  $10 \log_{10}$

## Exercise 7 — Adding powers that are given in dB

---

### Task

A receiver receives a power of **35 dBm** each from two different transmitters. Calculate the **total** received power in dBm.

(The power is received *incoherently*, i.e. the power contributions add up.)

$$35 \text{ dBm} + 35 \text{ dBm} \neq 70 \text{ dBm}$$

Convert to linear, add, convert back:

$$P_{\text{tot}} = 10 \log_{10} \left( 10^{\frac{35}{10}} + 10^{\frac{35}{10}} \right)$$

*Übungsblatt Wellenausbreitung, Aufgabe 7*

# EIRP — Effective Isotropically Radiated Power

- The Friis equation contains the product  $P_{Tx} \cdot G_{Tx}$
- Definition:

$$EIRP = P_{Tx} \cdot G_{Tx}$$

$P_{Tx}$ : radiated power of the transmitter

$G_{Tx}$ : antenna gain of the transmitter

- **Caution:** the radiated power is *not* identical with the output power of the transmit amplifier — e.g. because of cable losses between the amplifier output and the antenna input:

$$EIRP = L \cdot \underbrace{P_{ampl,out}}_{\text{with } P_{Tx}=L \cdot P_{ampl,out}} \cdot G_{Tx}$$

$L = 0 \dots 1$  (loss factor), often given in dB:

$$L_{dB} = 10 \log_{10} L \quad \Longleftrightarrow \quad L = 10^{\frac{L_{dB}}{10}}$$

One says “a loss of 5 dB” (*Verlust von 5 dB*), which means  $L_{dB} = -5$ .

# Exercises 3 and 4 — Radiated power and EIRP

---

## Exercise 3 — radiated power

An amplifier has an output power of **20 W**. The losses between amplifier output and antenna input are **2 dB**. What is the power radiated by the antenna, *in Watt*?

$$P_{Tx} = L \cdot P_{\text{ampl},out}$$

$$EIRP = P_{Tx} \cdot G_{Tx}$$

$$EIRP[\text{dBW}] = P_{\text{ampl},out}[\text{dBW}] + L_{dB} + G_{Tx}[\text{dBi}]$$

## Exercise 4 — EIRP

A base station uses an amplifier with an output power of **20 W**. The losses between amplifier and antenna are **3 dB**. The antenna gain is **18 dBi**. Calculate the **EIRP** in dBW and in dBm.

*Übungsblatt Wellenausbreitung, Aufgaben 3 und 4*

---

## Link Budgets

---



# Link Budgets (1)

---

- A **link budget** (also *link balance*) lists all necessary parameters — and their assumed values — that ensure the signal sent over a radio link exceeds a **minimum signal quality** at the receiver. The excess is the reserve = **margin**.
- A minimum signal quality may be given, for example, by a minimum received power, a minimum signal-to-noise ratio (SNR), or a bit error rate.
- The link budget includes parameters of the base station and of the mobile station, such as
  - ▶ output power of the transmit amplifier
  - ▶ losses between the transmit amplifier and the antenna
  - ▶ antenna gain of the transmitting antenna
  - ▶ antenna gain of the receiving antenna
  - ▶ etc.

Reference: T. Kürner, *Planung terrestrischer Netze*, TU Braunschweig, 2009.

# Exemplary link budget for GSM1800

$$\text{Uplink: } +30 - 3 + 18 + 5 - 2 - 6 - 3 = 39; \quad 39 - x = -104 \Rightarrow x = 143 \text{ dB}$$

- Beispiel: Linkbilanz für GSM1800 (GSM 03.30 Version 8.3.0. Release 1999, ETSI TR 101 362 V8.3.0) für Mobile-Klasse 1 (1W Ausgangsleistung) zuzüglich der Verluste Antenne-Mensch

|                                                                               | UL   | DL   |                                                        |
|-------------------------------------------------------------------------------|------|------|--------------------------------------------------------|
| + maximale Sendeleistung [dBm] Max Tx power                                   | 30   | 42   | Sender (Tx)<br>Transmitter                             |
| - Leitungsverluste [dB] Transmission line losses (e.g. cable)                 | 0    | 2    |                                                        |
| - Tx Losses Verluste Sender (Isolation, Combiner, Filter)                     | 0    | 3    |                                                        |
| + Sendeantennengewinn dBi Tx antenna gain                                     | 0    | 18   |                                                        |
| - Verluste Antenne-Mensch [dB] Losses antenna-human                           | 3    | 0    | Empfänger (Rx)<br>Receiver                             |
| - minimale Empfangsleistung [dBm] Min Rx power                                | -104 | -100 |                                                        |
| + Empfangsantennengewinn dBi Rx antenna gain                                  | 18   | 0    |                                                        |
| + Diversity-Gewinn [dB] Diversity gain                                        | 5    | 0    |                                                        |
| - Verluste Antenne-Mensch [dB] Losses antenna-human                           | 0    | 3    | UL: MS emits; BS receives<br>DL: BS emits; MS receives |
| - Leitungsverluste [dB] Transmission line losses (e.g. cable)                 | 2    | 0    |                                                        |
| - Fading reserve Schwundreserve ( $P_{fo} = 75\%, \sigma = 8\text{dB}$ ) [dB] | 6    | 6    |                                                        |
| - Interferenzsicherheitsreserve [dB]                                          | 3    | 3    |                                                        |
| = maximale Ausbreitungsdämpfung [dB] Max. propagation loss [attenuation]      | 143  | 143  |                                                        |

UL: MS transmits, BS receives

DL: BS transmits, MS receives

The calculated max. propagation loss determines — depending on the topography — the **cell radius**.

Reference: T. Kürner, *Planung terrestrischer Netze*, TU Braunschweig, 2009.

## Exercise 6 — Path loss models and the cell radius

### Task

The minimum reception power for maintaining a radio connection can be described by

$$P_{Rx,min}[W] = C \cdot \left( \frac{1}{r[m]} \right)^n$$

with  $C$  a constant,  $r$  the Tx–Rx distance and  $n$  the **path loss exponent**.

In a propagation situation modelled with  $n = 2$  the cell radius is **1 km** ( $P_{Rx,min}$  is reached at the cell edge). The same base station, with the same transmit power, is now used in a scenario with  $n = 3$ . Calculate the new cell radius.

Free space  $n = 2$  | two-path, beyond  $d_{BP}$ :  $n \approx 4$  | urban NLOS:  $n \approx 3 \dots 5$

*Übungsblatt Wellenausbreitung, Aufgabe 6*

# Interactive lab 4 — Rural macrocell path loss (3GPP TR 38.901)

---

## What you can explore

The *real* path-loss model that network planners use — the 3GPP Rural Macrocell model, LOS and NLOS — instead of our idealised exponent  $n$ . Sliders for carrier frequency, antenna heights, building height and street width.

### Try this:

- Read the slope before and after the breakpoint  $d_{BP}$  and compare with the exponent  $n$  of Exercise in slide 55.
- Enter the **143 dB** from the GSM1800 link budget (slide 54) — which distance does the model give? That is your cell radius.
- Switch LOS → NLOS at fixed distance: how many dB does a lost line of sight cost you?



[Colab notebook 05](#)

[3GPP RMa path loss](#)

Runs in Google Colab  
— no installation needed.

# Link Budgets: maximum permissible propagation attenuation (1)

---

From

$$P_{Rx} = P_{Tx} \cdot G_{Tx} \cdot \underbrace{L_{Funkkanal}}_{\substack{<1 \\ \text{propagation loss,} \\ \text{e.g. free space}}} \cdot \underbrace{G_{sonstiges}}_{\substack{>1 \\ \text{other} \\ \text{gains}}} \cdot \underbrace{L_{Sonstiges}}_{\substack{<1 \\ \text{other} \\ \text{losses}}} \cdot G_{Rx} \quad \text{follows}$$

$$L_{Funkkanal} = \frac{P_{Rx}}{P_{Tx} \cdot G_{Tx} \cdot G_{sonstiges} \cdot L_{Sonstiges} \cdot G_{Rx}} \quad (1)$$

$$L_{Funkkanal}[\text{dB}] = 10 \log_{10} L_{Funkkanal} < 0 \quad \text{e.g. } -100 \text{ dB}$$

The attenuation of the radio channel is often given as a **positive** value (e.g. “100 dB attenuation”), i.e. one calculates  $10 \log_{10}(1/L_{Funkkanal}) = -10 \log_{10} L_{Funkkanal} = -L_{Funkkanal}[\text{dB}] > 0$ .

# Link Budgets: maximum permissible propagation attenuation (2)

---

Reciprocal value (*Kehrwert*) of (1) gives:

$$\frac{1}{L_{\text{Funkkanal}}} = \frac{P_{Tx} \cdot G_{Tx} \cdot G_{\text{sonstiges}} \cdot L_{\text{Sonstiges}} \cdot G_{Rx}}{P_{Rx}}$$

(Positive) propagation loss (*Funkfelddämpfung*) in dB:

$$\begin{aligned} L_{\text{Funkkanal},\text{pos}}[\text{dB}] &= 10 \log_{10} \left( \frac{1}{L_{\text{Funkkanal}}} \right) \\ &= P_{Tx}[\text{dBm}] + G_{Tx}[\text{dBi}] + \underbrace{10 \log_{10} G_{\text{sonstiges}}}_{>0} + \underbrace{10 \log_{10} L_{\text{Sonstiges}}}_{<0} + G_{Rx}[\text{dBi}] - P_{Rx}[\text{dBm}] \end{aligned}$$

If the “other losses” are given as a **positive** dB number, they must be **subtracted**, so that the equation above is fulfilled. Then:

$$\begin{aligned} L_{\text{Funkkanal},\text{pos}}[\text{dB}] &= P_{Tx}[\text{dBm}] + G_{Tx}[\text{dBi}] + \text{other\_gains}[\text{dB}] \\ &\quad - \text{other\_losses}[\text{dB}] + G_{Rx}[\text{dBi}] - P_{Rx}[\text{dBm}] \end{aligned} \quad (2)$$

# Link Budgets: maximum permissible propagation attenuation (3)

---

- Equation (2) describes the **maximum propagation loss** that is allowed in order to maintain a given (minimum) received power  $P_{Rx}$ .
- In a link budget a **safety margin** is usually introduced — e.g. one wants to stay 3 dB above the minimum received power.
  - ▶ then formula (2) is used, the margin (in dB) is subtracted, and a correspondingly **smaller** maximum propagation loss is obtained

Smaller permissible propagation loss  $\Rightarrow$  smaller cell radius  $\Rightarrow$  more base stations.

---

Review questions

*Fragen zu Foliensatz 1*

---



# Review questions 1–4 (*Fragen 1–4*)

---

1

*Was versteht man unter Mehrwegeausbreitung?*

What is meant by multipath propagation?

→ slide 14

2

*Welche Faktoren kann man zu Link Budgets zählen?*

Which factors belong in a link budget?

→ slides 53, 54

3

*Unterschied fast bzw. slow fading?*

What is the difference between fast and slow fading?

→ slides 27, 28

4

*Was sagt die Friis-Gleichung aus?*

What does the Friis equation tell us?

→ slides 17, 19

# Review questions 5–8 (*Fragen 5–8*)

---

5

*Mit welchem Kriterium kann man Streuungseffekte vernachlässigen?*

With which criterion can scattering effects be neglected?

→ slides 23, 29

6

*Ab wann müssen Hindernisse in den Funkweg einberechnet werden?*

From when on must obstacles be taken into account?

→ slides 40, 41

7

*Was ist der Knife-Edge-Ansatz?*

What is the knife-edge approach?

→ slides 36, 37

8

*Was ist Fading?*

What is fading?

→ slides 24, 26

# Where to practise — exercises and interactive labs

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## Übungsblatt Wellenausbreitung

| Exercise                            | Slide |
|-------------------------------------|-------|
| 1, 2, 5 — dBm, mW, the 30 dB rule   | 47    |
| 3, 4 — radiated power, EIRP         | 51    |
| 6 — path loss exponent, cell radius | 55    |
| 7 — adding powers in dB             | 49    |
| 8 — free space attenuation          | 18    |
| 9 — knife-edge diffraction          | 37    |

Solutions: *Lösungsblatt Wellenausbreitung*.

## Interactive Colab notebooks

| Notebook                  | Slide |
|---------------------------|-------|
| 01 — Multipath & Friis    | 15    |
| 03 — Reflection, two-path | 25    |
| 02 — Huygens, diffraction | 33    |
| 05 — 3GPP RMa path loss   | 56    |



[github.com/oalnaseri/mobilcom\\_course](https://github.com/oalnaseri/mobilcom_course)